

Shared representations, practical adapters

A bounded test of PRH and two engineering implementations

Experiment A Experiment B Shared hub

PRH = Platonic Representation Hypothesis · 20-minute talk

Research question and practical constraints

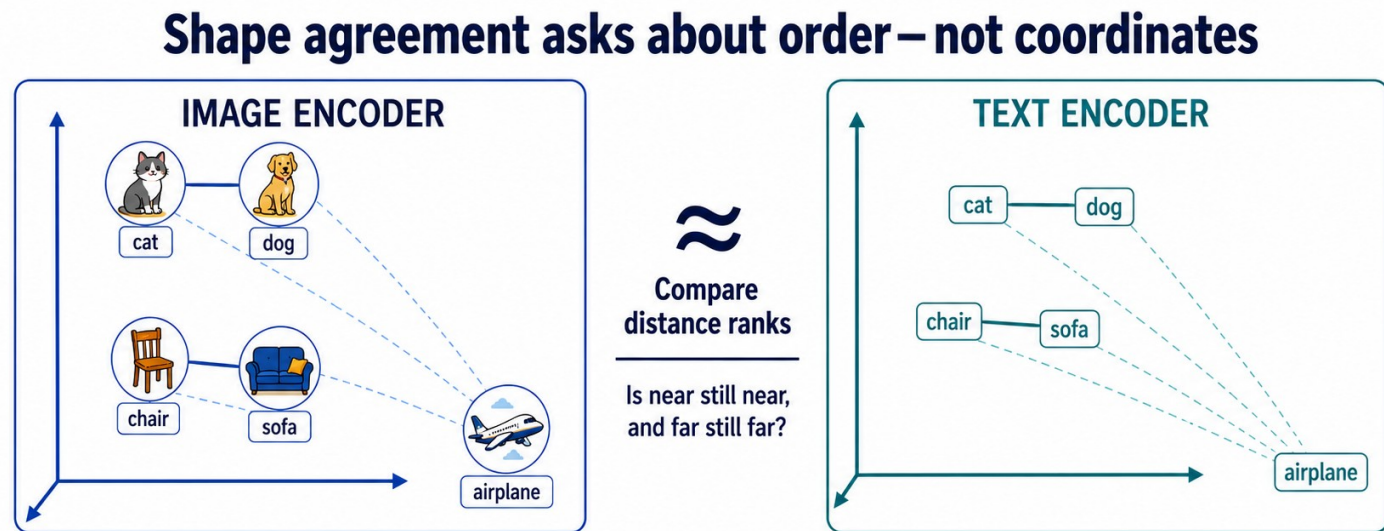
Scientific scope: shared relations and useful linear mappings on held-out items.

Practical constraint: limited compute and device memory favor frozen, smaller encoders.

Engineering goal: reuse a stronger search space while a weaker device runs its own encoder.

Supported scope: selected models, datasets and metrics. Universal convergence remains open.

Modality, objective and representation space



Modality = pixels or text

Objective = what training rewards

Embedding = numerical representation

Structure = relations among items

Shared structure means similar relationships among items.

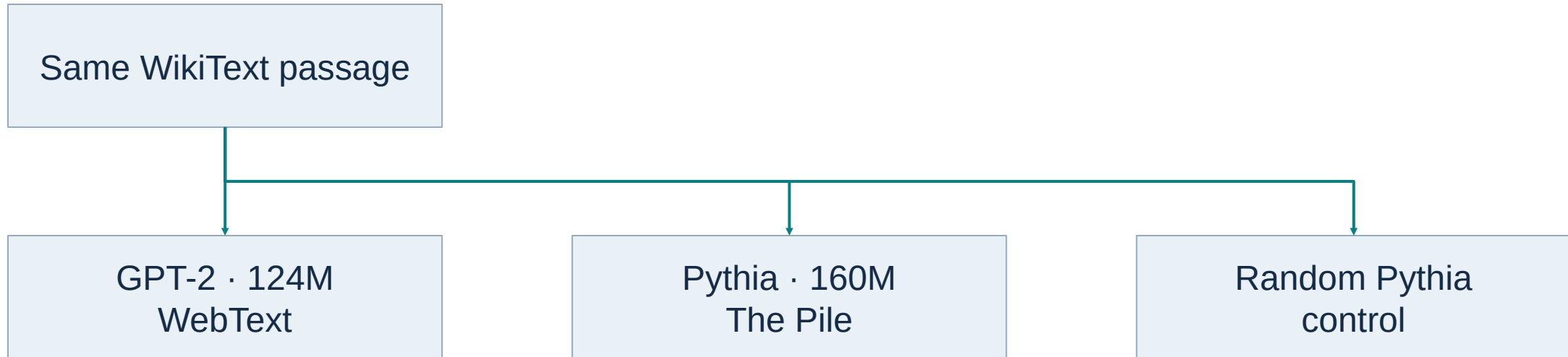
Cat–dog diagrams are explanatory schematics, not measured embeddings.

Data and what this project actually fits

Study	Data source	Fit / evaluation	Learned in this project
A	WikiText-103: Wikipedia	7,500 / 2,500 passages	Layer-wise linear maps
B	COCO val2017	3,000 / 1,000 images	512 × 768 adapter
Hub	COCO train2017 caches	8,533 / 1,000 items	PCA, entry maps, caption head
F follow-up	Teacher image embeddings	37,999 / 2,000 rows	Ridge and separate MLP

B uses one caption per image. Hub targets average five caption embeddings.

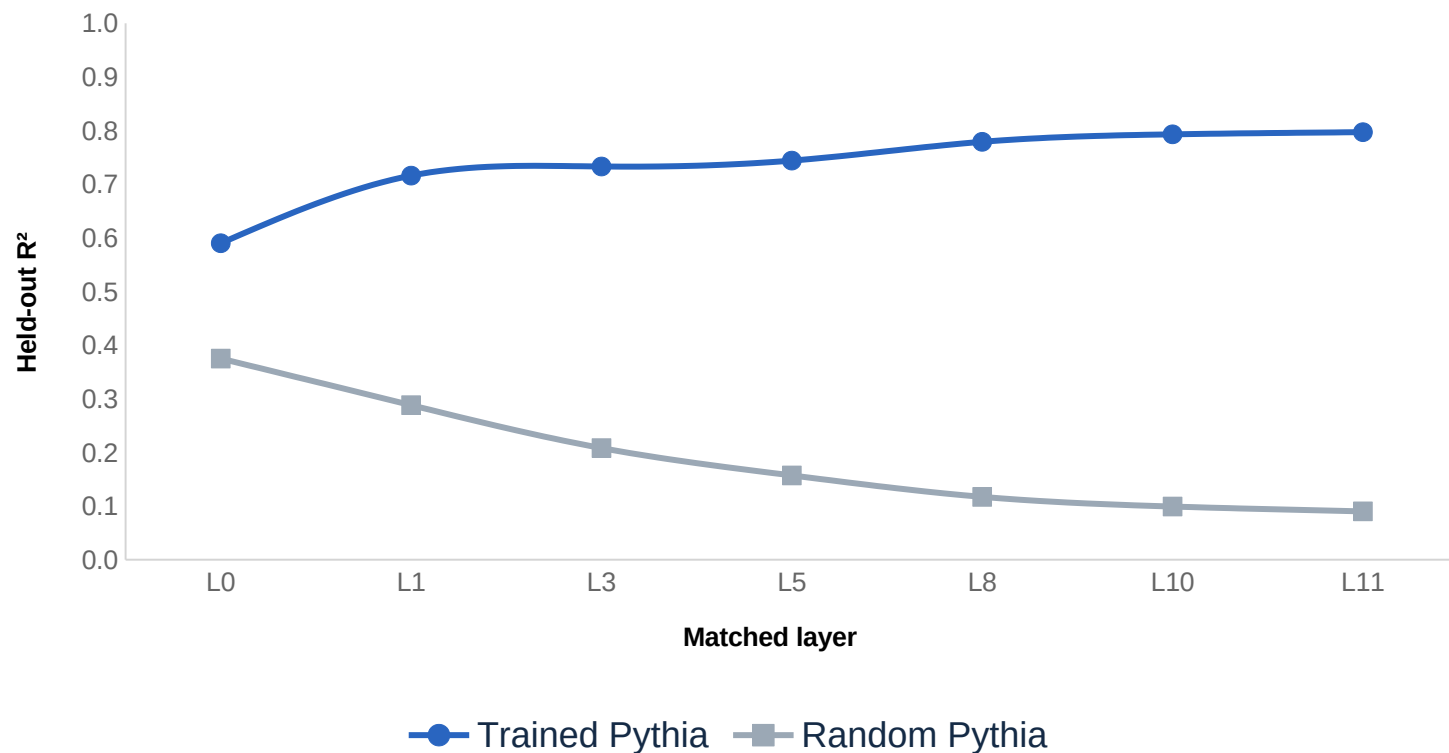
Experiment A: independent language models



13 states per model, each pooled to 768 coordinates

CKA (centered kernel alignment) compares relational structure. Ridge predicts coordinates.

Experiment A: learned structure exceeds control



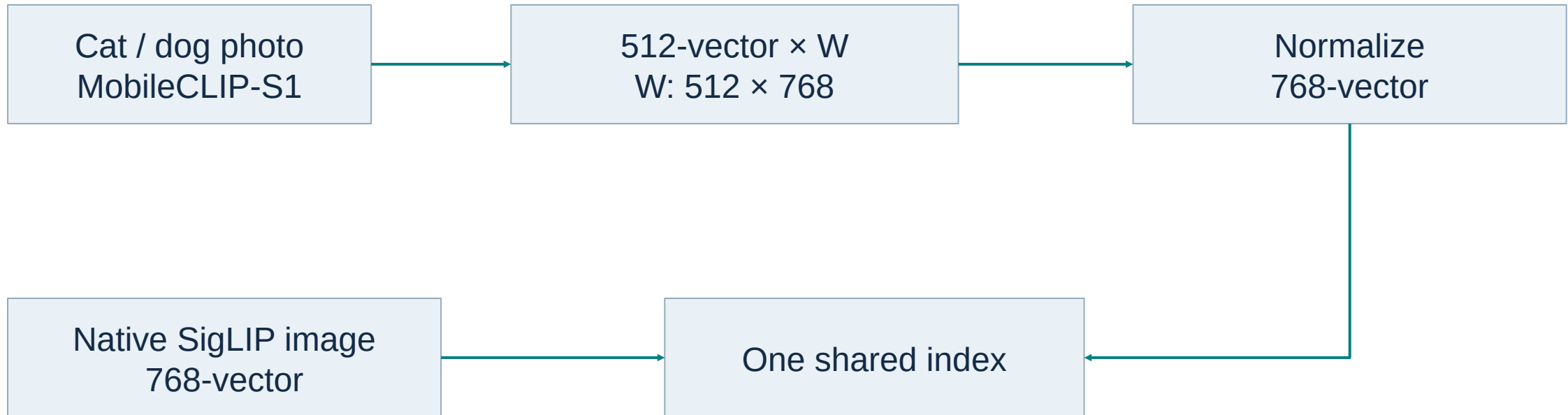
$$R^2 = 0.797$$

At L11, a linear map explains about 80% of target variance.

CKA: similarity-pattern agreement
Mean 0.359 trained
Mean 0.067 random

Measured alignment supports the restricted hypothesis on this model pair.

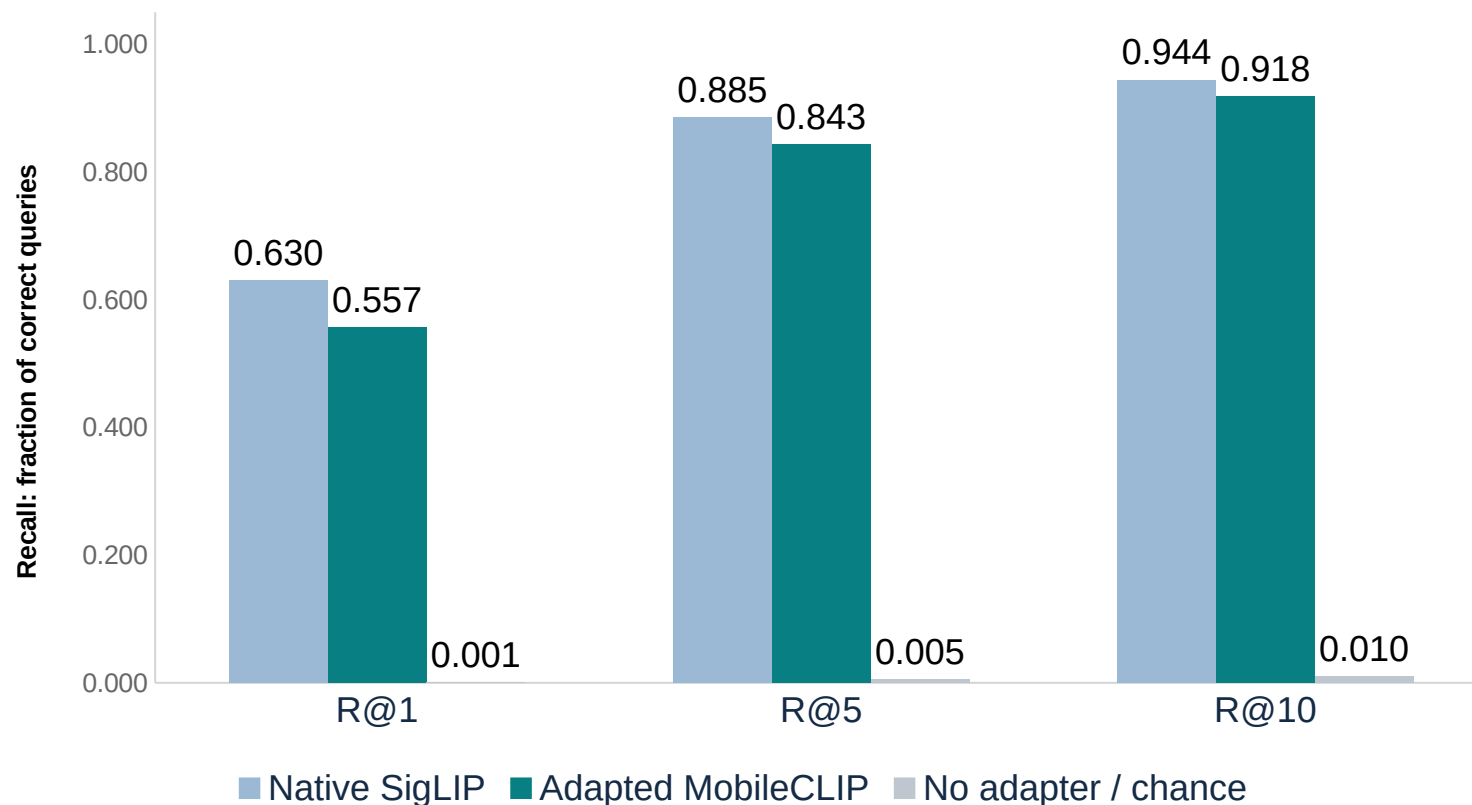
Experiment B: one index for phone and server



Query “a cat on a chair”: SigLIP text supplies the comparison vector.

Fit: 3,000 paired images. Test: 1,000 held-out images.

Experiment B: retrieval quality and its cost



95.3%

of native R@5
0.843 / 0.885

R@1 retention: 88.4%
Below the 90% gate

R@5: the matching image appears among the first five results.

Small adapter, measurable deployment advantages

0.79 MB fp16 adapter: 393,216 coefficients; one matrix multiply plus normalization.

MobileCLIP-S1 vision tower: 21.5M parameters. Small enough to target constrained devices.

Core ML export and shared Qdrant index: reuse one search space across device tiers.

Cosine measures direction agreement: MLP 0.903 versus ridge 0.900. Little measured gain.

Reported prototype timings are not device-wide latency, battery or privacy guarantees.

Cross-modal test before the hub

Image: DINOv2 representation. Text: bge representation of matching captions.

Fit paired vectors with ridge while both pretrained encoders stay frozen.

$R@1 = 0.358$: the correct target ranks first for 35.8% of 1,000 test queries.

40.9% of matched reference

0.358 / 0.875. Five-caption averaging applies to both method and reference.

Engineering system 2: the shared hub

Configuration	Members defining the basis	Concat width
Published 4-space	DINO small/base/large + bge	5,376
Intermediate 5-space	Four-space + GPT-2	6,144
Extended 7-space	Five-space + BERT + SBERT	7,680

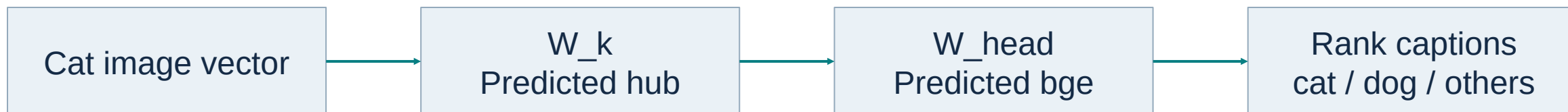
Scale each space · concatenate · PCA to 512 · whiten

Reference H comes from the fitted construction. Entry maps then learn to predict H.

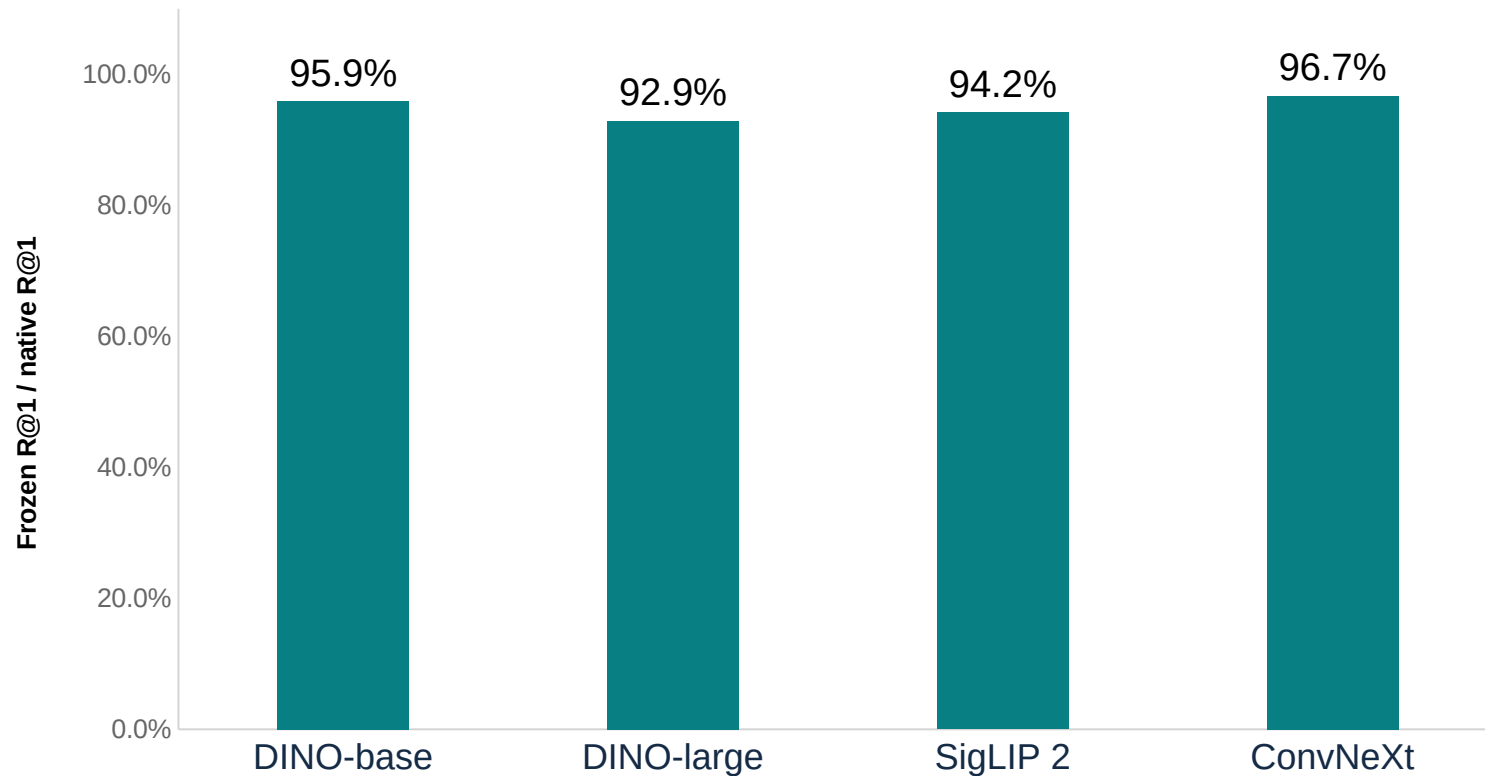
Training references and prediction routes

Fit	Input	Training reference / target
Hub basis	Aligned member embeddings	Data-derived PCA coordinates H
Entry map W_k	Encoder features X_k	Reference hub coordinates H
Caption head W_{head}	$H_{small} = X_{small} W_{small}$	Actual bge caption vectors T_{bge}

At prediction time



A frozen head transfers to new encoders



ConvNeXt: 96.7%

One new entry map.
Same frozen hub.
Same frozen caption
head.

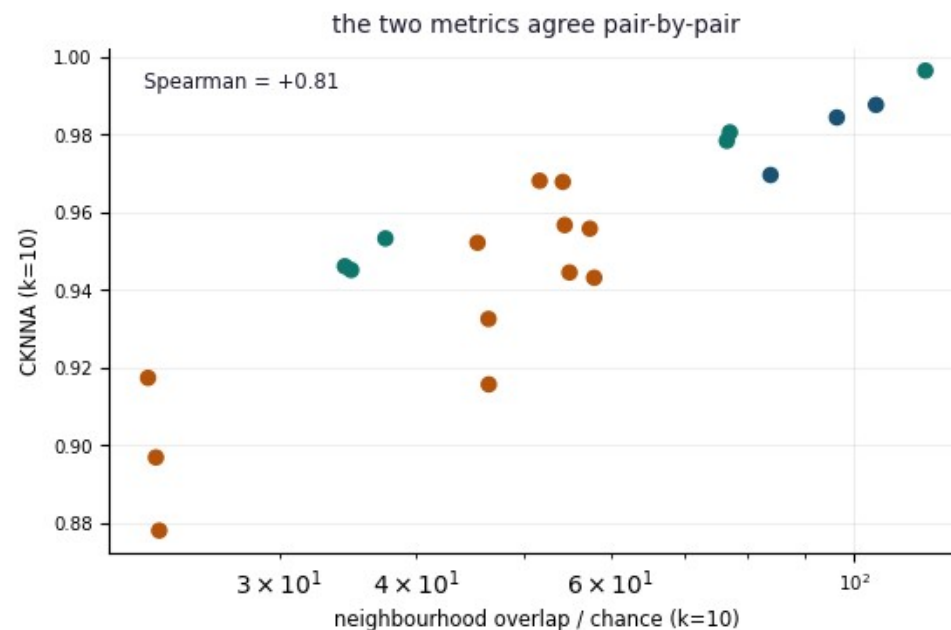
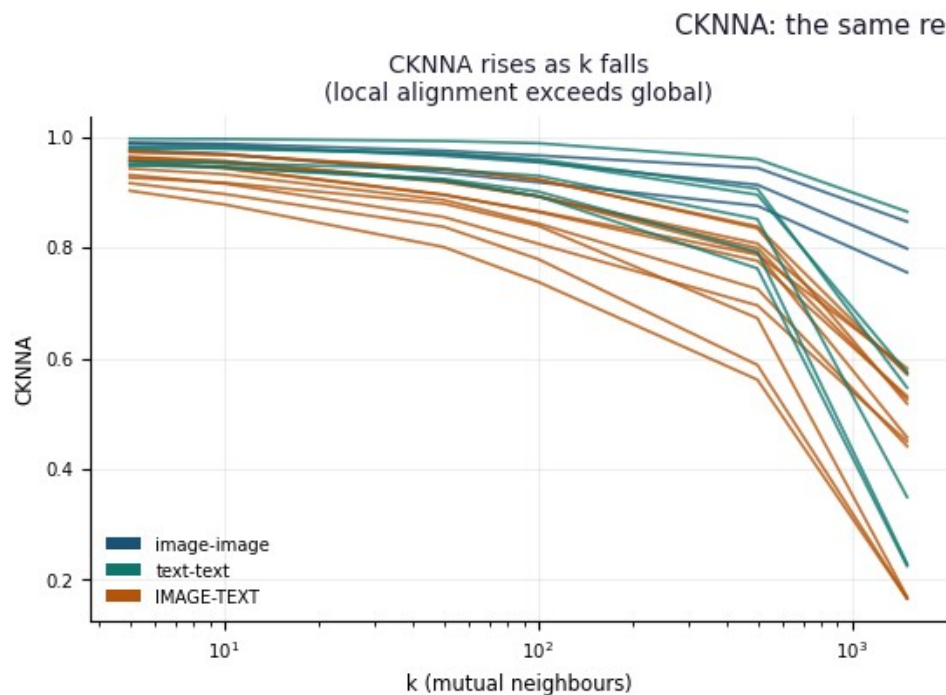
Retention measures reusable-head performance relative to an encoder-specific head.

Similarity has several meanings

Measurement	Cat-dog question	Map fitted?
Cosine	Do these compatible vectors point alike?	No, or after mapping
Spearman ρ	Do both spaces rank item distances alike?	No
k-NN overlap	Does this cat have the same neighbors?	No
CKA / CKNNA	Do similarity patterns agree globally / locally?	No
Ridge / Procrustes	Does a full linear map / rotation recover targets?	Yes

CKA = centered kernel alignment. CKNNA measures alignment of mutual-neighbor structure.

The strongest shared structure is local



21 / 21 pairs: local CKNNA exceeds the global limit

CKNNA compares local similarity structure. The falling overlap/chance curve alone is insufficient.

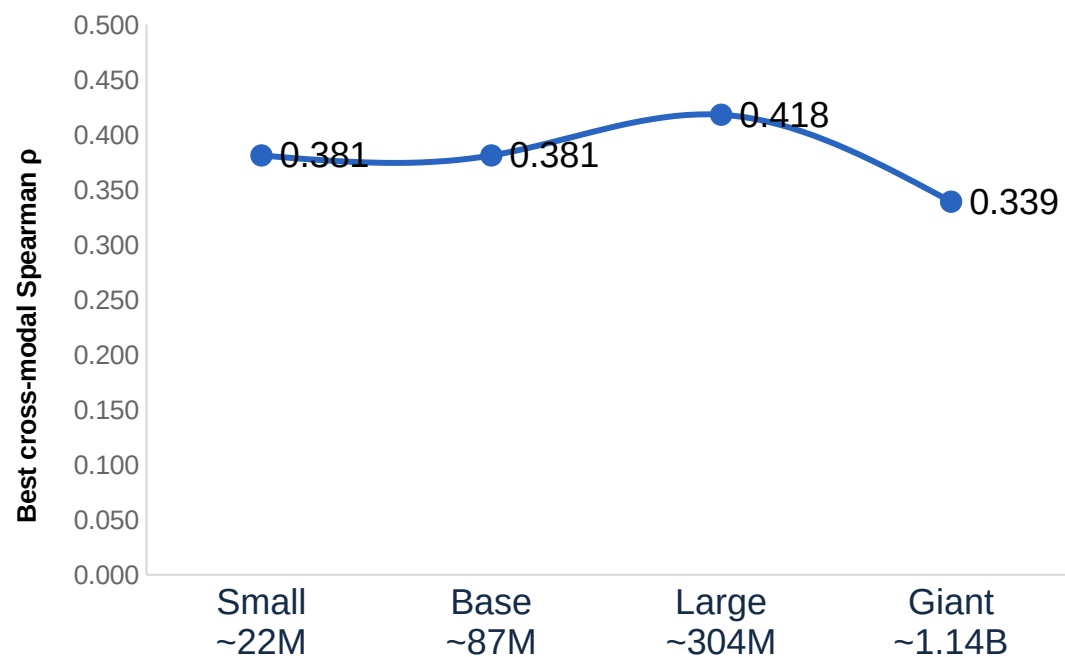
GPT-2 separates reading from writing

Space	Read out	Write into	Interpretation
GPT-2	95.1%	24.8%	Recoverable content, fragile target ranking
bge-m3	98.1%*	98.1%	Useful caption target
SBERT	Not supplied	101.0%	Relative target score, not accuracy

Read: source vector into hub. Write: hub prediction into a target space.

***bge read figure comes from the supplied discussion; diagnostics use their own references.**

Larger-model tests: progress and limits



ρ ranks pairwise distances with no fitted map. Bigger did not mean higher agreement here.

Six controls and the numbers they guard

Control	Fitted result	Control / reference	What this checks
Random map / no adapter	AUC .999 G4 R@1 .557	AUC .477 G4 R@1 .003	The learned bridge supplies useful information.
Random model weights	Mean CKA .359 Mean R ² .737	Mean CKA .067 Mean R ² .173	Learned representations exceed the random-weight comparison.
Shuffle item pairing	E1 R ² .511	Shuffled R ² -.130 Gap .641	Correct item correspondence matters to the fitted map.
Matched native reference	B R@5 .843	Native R@5 .885 95.3% retention	Task, rows and caption treatment must match.
Held-out ConvNeXt	R@1 retention 96.7%	ViT band 93.8–96.5%	Tests a supervised convnet outside the hub membership.
Held-out SigLIP 2	R@1 retention 94.2%	Encoder-specific head	Tests an unrelated lineage with the caption head frozen.

R² explains target variance. AUC separates pairs. R@K counts successful retrievals.

Failures sharpened the claims

Observation	Lesson / correction
A: initial fit underpowered	Increase paired rows before interpreting a failure.
B: MLP did not improve R@1	A linear map was sufficient in the tested controls.
B: 88.4% retention at R@1	Report the missed 90% gate.
Hub: width cliff	Near-null head-fit direction; regularization or truncation repairs it.
Metric traps	Ratios, geometry and chance need matched interpretation.

Next tests, in priority order

Priority	Test	Success evidence
1	Expand paired corpus; repeat giant / Qwen transfer	Stable absolute R@K, intervals and non-degenerate references
2	Extend model scale with matched pooling and controls	Raw geometry and transfer reported separately
3	Small experts versus a large encoder	Matched task quality, device latency, RAM and energy
4	Domains, languages, image-target heads and mobile tests	Transfer beyond COCO; actual device benchmarks

Completed: initial larger-model geometry tests. Pending: stronger power and broader validation.

Achievements and the defensible conclusion

Science: shared relations and linearly recoverable information on held-out data.

Engineering 1: 0.79 MB adapter; 95.3% of native retrieval at R@5.

Engineering 2: reusable hub head; ConvNeXt retains 96.7% of its native reference.

Boundary: useful interoperability supports a restricted PRH claim. Broader proof remains open.

Questions · experiment inventory, model details, terms and resources in the appendix

Appendix: model inventory, language and text

Model	Approx. parameters	Vector width	Pretraining / objective	Role
GPT-2	124M	768	Next-token, self-supervised; WebText	A; hub text source
Pythia	160M	768 per layer	Next-token, self-supervised; The Pile	A counterpart/control
BERT-base	~110M	768	Masked-token + original NSP objectives	Text diagnostic; 7-space
SBERT all-mpnet	~110M	768	MPNet + contrastive sentence-pair fine-tuning	Text diagnostic; 7-space
bge-m3	~568M	1,024	Multilingual retrieval, contrastive and unified tuning	Hub member and target
Qwen3 Embedding	0.6B / 4B	1,024 / 2,560 (max.)	Embedding-tuned causal model; last-token pooling	Scale extension

Appendix: model inventory, vision

Model	Approx. size	Stored width	Training signal	Role
MobileCLIP-S1	21.5M vision	512	Image-text pairing + distillation	Phone encoder, B
SigLIP 2 base	~0.4B full model	768	Sigmoid image-text loss + auxiliary objectives	B target; held-out hub route
DINOv2 S / B / L	~22 / 87 / 304M	768 / 1536 / 2048	Self-supervised DINO + iBOT; no caption loss	E ladder; hub members
DINOv2 giant	~1.14B	3,072 CLS+patch	Same visual family; matched pooling	Held-out scale test
ConvNeXt-base	~89M	1,024	ImageNet-22k class supervision	Held-out convolutional route

Appendix: supervision and geometry terms

Term	Meaning in this project
Self-supervised	Targets come from the data itself: next token, masked token, image views.
Supervised / pair-supervised	Known labels or correspondences guide fitting; ridge pairs also count.
Unsupervised matching	Recover correspondence from geometry without supplied item pairs.
PCA / whitening	Choose variance directions / rescale retained directions.
Anisotropy / effective rank	Uneven directional statistics / spectral concentration, not literal semantic dimensions.
Manifold / coordinate chart	A low-dimensional geometric model / a representation of it; an interpretation here.

Appendix: metric definitions and blind spots

Metric	Meaning	Caution
Cosine	$x \cdot y / (x y)$: direction agreement	Near-one scores can be uninformative in collapsed spaces.
R^2	$1 - SSE/SST$: target variance explained	Includes magnitude; may be negative.
Spearman ρ	Correlation of ranked within-space pair distances	No map needed; not retrieval accuracy.
CKA	Agreement of centered similarity structures	Calibrate against controls and width effects.
CKNNA	Centered alignment on mutual k-nearest neighbors	Specify k; compare local and global within this metric.
k-NN overlap	Shared neighbors divided by k	Null $\approx k/(N-1)$ when self is excluded.

Appendix: retrieval, uncertainty and power

Metric	Reported example	One-sentence reading
R@K	B R@5 = 0.843	Correct image appears among first five for 84.3% of queries.
Retention	B 95.3%; ConvNeXt 96.7%	Ratio to the appropriate native reference within each protocol.
Verification AUC	0.999	Probability of ordering a positive pair above a sampled negative.
Instance ranking R@1	0.987	Match the same photo across encoders, not a caption query.
Membership AUC	0.889	Ranks member queries above nonmembers across score thresholds.
Power / uncertainty	Giant: 2.8 rows/dim	Insufficient fitting evidence under the project's prespecified heuristic.

Appendix: experiment inventory A and B

ID	Test / implementation	Reported outcome / status
A1	Extract all layer representations on WikiText	10k passages; trained and random models
A2	CKA, layer matching and geometry diagnostics	Trained/control gap; later hub-dominance caveat
A3	Ridge stitching; small versus larger sample run	Mean held-out R^2 0.737; peak 0.797
B1–B3	Paired caches; ridge/Procrustes/MLP; retrieval	Ridge cosine .900; R@1 .557; R@5 .843
B4–B5	Core ML export; fp16 parity; qualitative galleries	0.79 MB artifact; held-out qualitative demonstrations
B6–B10 / D	Live demo, quantization and decision diagnostics	Reported complete; stage numbering varies by document

Appendix: experiment inventory C, E and F

ID	Question	Result / status
C1 / C1.1	Recover pairs without correspondence labels?	GW exact match 39.7% vs 1/1500 chance; ICP and whitening hurt.
E1 + sweep	DINO image to bge text, and S/B/L capacity	40.9 / 53.5 / 58.3% of matched R@1 reference.
E1.1–E1.3	GPT-2 swap, whitening rescue, text ladder	Geometry affects retrieval; pre-registered rank ordering supported.
E2 / E3	Search demo and functional consumer test	Demo built; R@5 stitching penalty .049; internal splice scaffolded.
F1–F2 / F5	Teacher corpus, MLP versus ridge, larger refit	No nonlinear gain; refit R@1 +.0244 in paired evaluation.
F3–F4	Unfreeze student trunk / change bottleneck	Built but deliberately gated and unrun.

Appendix: experiment inventory G0–G7

ID	Question	Reported result
G0-exact	Recover published protocol	Four members, scalar scaling, seed-0 split; SigLIP 94.2% reproduced.
G1 / G2	Shared hub, composition, working demo	Seven-space R@1 .451 vs five-space .445; live retrieval prototype.
G3	Shared versus bespoke pairwise hubs	42 ordered pairs; mean relative performance 103.6%.
G4 / G4-conv	Held-out lineage and architecture	SigLIP 94.2%; ConvNeXt 96.7% of native in published comparison.
G5 / G6	Scale of agreement; manifold diagnostics	21/21 local CKNNA stronger; mean R ² : ridge .667 vs Procrustes .194.
G7	Local-structure effect and width sweep	Healthy-pair effect near zero; width cliff required later diagnosis.

Appendix: experiment inventory G8–G16, K, H, V

ID	Test	Outcome / status
G8	Four candidate discontinuity statistics	All failed to track the observed transfer cliff.
G9 / G11 / G12	Source dimension and target swaps	Cliff location followed source width in tested conditions.
G10	Hub text composition and target overlap	Member identity matters; matching-target benefit disclosed.
G13–G16	Ablations, rank, singular direction and repair	Corrected flawed tests; truncation / stronger head ridge repairs fit.
K1–K3	Raw Procrustes, reproduction, bounded null	Healthy-pair gain CI [–.038, +.024]; prespecified $\pm$.050 margin.
H1 / V1	Artifact, power, shuffle and independent checks	47 checks, no hard failures; verification AUC .9994 reproduced.

Appendix: scale tests and remaining tasks

Test	Observed / planned	Interpretation
P1: giant transfer	100% ratio; 2.8 fitting rows/dim	Inconclusive, reference degeneracy.
P2: giant raw ρ	.339 versus large .418	Tested monotonic threshold failed.
P3: Qwen-4B raw ρ	.307 versus bge .320	Tested threshold failed; model-family confounds remain.
P3b / P4	.6B ρ .352; geometry and transfer diverge	Architecture control; do not infer transfer from ρ .
Next: more paired rows	35,133 total / 4,000 eval proposed	31,133 fit rows; ~10.1 per giant input dimension.
Next: new studies	Experts vs generalist, domains, device power	Prospective tasks; no claimed outcomes yet.

Appendix: the map W and linear controls

W is the learned translation matrix. Prediction: $x [1 \times d] \cdot W [d \times p] = \hat{y} [1 \times p]$.

Ridge fit: $W = (X^T X + \alpha I)^{-1} X^T Y$. α controls the penalty on large coefficients.

SVD: $W = U \Sigma V^T$. Orthogonal transforms and directional scaling describe the map.

Experiment B control	R ² : target variance explained	Cosine: direction agreement
Ridge, full linear map	0.592	0.900
Procrustes, orthogonal alignment	0.038	0.775

Appendix: hub equations and training roles

Hub construction: $H = \text{whiten}(\text{PCA}_{512}(\text{concat}(X_1/s_1, \dots, X_K/s_K)))$

Entry fit: $W_k = \text{argmin}_W ||X_k W - H||^2_F + \alpha_k ||W||^2_F$

Head input: $H_{\text{small}} = X_{\text{small}} W_{\text{small}}$

Head fit: $W_{\text{head}} = \text{argmin}_W ||H_{\text{small}} W - T_{\text{bge}}||^2_F + \alpha_h ||W||^2_F$

Query: $\hat{t} = x_k W_k W_{\text{head}}$; rank actual bge captions by cosine.

Appendix: source disagreements resolved

Source wording / ambiguity	Treatment in this presentation
4, 5 or 7 members	Separate configurations: 5376, 6144 and 7680 concat dimensions.
Column z-score versus space scaling	Later E.14 audit specifies one scalar per space.
Small denominator is “conservative”	Unsupported: fixed-numerator ratios increase as denominator shrinks.
MLP equality proves exact linearity	Finite controls support linear sufficiency under tested conditions.
Status lists completed tests as TODO	Use later report addenda for completion status.
SBERT as NLI-only BERT	Use all-mpnet-base-v2 provenance from its official card.

Appendix: research and implementation resources

Resource	Use
Huh et al. (2024), PRH; Kornblith et al. (2019), CKA	Scientific motivation and representation metrics
COCO; Salesforce WikiText-103	Image-caption pairs and passage data
Apple MobileCLIP; Google SigLIP 2; Meta DINOv2	Official vision model code and provenance
EleutherAI Pythia; BAAI bge-m3; Sentence Transformers	Official language and text model cards
Final_Project_Report(7), Results_Summary(3), Project_Atlas(4)	Reported experiments, protocol details and later corrections
PyTorch, Transformers, scikit-learn, Core ML, Qdrant	Encoding, linear fitting, device export and shared retrieval

Full links and per-slide sources appear in speaker notes.